Environmental risk analyses for new materials and systems

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Plastics are widely used across all sectors of the economy, and production has been increasing since they first hit the market in the 1950s. Worldwide production recently topped 450 million metric tons with a market value exceeding $500 billion U.S. dollars, and continued growth is projected. Conventional polymers are made from petroleum, and alternate sources – notably biofeedstocks – are now being pursued to reduce associated greenhouse gas emissions. Having been designed to last, conventional plastics persist in the environment with half lives that can extend from decades to centuries. With a dismal recycling rate of 10-15%, the environmental accumulation of waste plastics is a growing global challenge. Furthermore, unintended consequences can be expensive. To illustrate, the liability cost in the United States for per- and polyfluoroalkyl substances – which only after production and widespread use were found to be highly mobile and persistent – could total $175 billion.

Innovative developers seeking to reduce the environmental footprint of new polymer products across their lifecycle and avoid downstream liabilities need a way to screen for both recyclability and potential end-of-life fate and effects during design. Predictive tools have been developed to address both aspects. For recyclability, the screening tool created to facilitate "design-for-recycling" couples classifier-derived sorting information (using a convolutional neural network approach) with multi-level statistical entropy analysis (MSEA) to quantify plastic recoverability under representative end-of-life scenarios. The resulting recyclability index provides a quantitative measure of how well a plastic product or material stream can retain value and remain recoverable over time. This recyclability index approach provides a quantitative basis for comparing plastic design choices, sorting technologies, and end-of-life management strategies. It can support design-for-recycling by identifying product attributes that improve material identification, reduce contamination, and preserve material value across life cycles. From a management and policy perspective, the tool can inform material labeling, digital product passports, extended producer responsibility programs, recycling infrastructure investment, and performance standards for recyclable plastics. More broadly, the coupled sorting-MSEA approach supports risk-relevant decisions by making uncertainty in recovery performance explicit and helping align plastic circularity goals with improved end-of-life management. While this tool was created to guide the development of more environmentally friendly plastics, the approach can be extended to inform the design of other materials to facilitate their end-of-life recyclability.

For the environmental risk aspect, a framework has been created to predict potential downstream fate and effects in the environment. These tools enable developers to screen for potential mobility, persistence, bioaccumulation, and toxicity of plastics over time. This environmental framework combines quantum chemistry (first principles) and machine learning techniques that incorporate data from the literature and other online datasets to predict end-of-life properties of plastics from structural characteristics. The components include predicting plastic degradation rates and lifetimes from a number of factors, including environmental compartment (e.g., water, air), glass transition temperature , and density, as well as predicting the adsorption coefficient of co-contaminants adsorbing to plastics by using a set of molecular descriptors for both the plastic and the co-contaminant.

The presentation will highlight the approaches used to develop predictive tools to screen candidates for new plastics (polymers and additives) during their design stage for both recyclability and end-of-life fate and effects. These tools can also be expanded to predict recyclability and properties that influence environmental fate and effects of other materials and systems, ranging from ion exchange resins and membranes for critical mineral separations to polymers used in a variety of energy system applications.

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